Application note 012d TVM

This application note aims to do the following:

- demonstrate R47/C47 Time Value of Money (TVM) through 10 worked examples.
- solve the TVM puzzles of Duncan's on your calculator by R47/C47 RPN programs.
- document the TVM formulas in use in the R47/C47
- demonstrate the comparative performance of the R47/C47 models

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Introduction

The R47/C47 TVM implementation follows HP financial calculator conventions but requires explicit setup of payment and compounding periods. The five core TVM registers are n (number of periods), I/a (annual interest rate as percentage), PV (present value), PMT (payment per period), and FV (future value).

Special settings control pp/a (payment periods per year), cp/a (compounding periods per year), and Begin/End mode (when interest applies). Functions EFF/a->I/a and I/a->EFF/a convert between nominal and effective annual rates.

Learning outcomes of tutorial problems included:

Section 3: 10 problems from a worked out financial calculator tutorial.

The learning outcomes of Section 3 focus on TVM fundamentals from the Baker & Lin tutorial: understanding how the five core variables interact, setting payment and compounding periods correctly, and recognizing when to use Begin versus End mode.

Examples 3A/3B/3C demonstrate different problem approaches.
Example 5 teaches period conversion (annual to bi-weekly).
Examples 6 and 7 show how adding constraints changes the calculation.
Example 11 covers nominal to effective interest rate conversions.

- Example 1: Calculate FV given initial investment and time horizon
- Example 2: Calculate PV of a future amount
- Example 3: Compare lottery options (lump sum vs annuity stream)
- Example 3A: Same comparison using BEGIN mode
- Example 3B/C: Alternative approach comparing future values instead of present values
- Example 4: Calculate required annual savings to reach investment goal
- Example 5: Calculate bi-weekly savings (26 periods per year)
- Example 6: Determine required rate of return given savings capacity
- Example 7: Required return with existing capital and ongoing contributions
- Example 11: Convert nominal to effective interest rates for quarterly and daily compounding

Section 4: 1 Worked out TVM AMORT example from an HP tutorial.

- Example 1: Calculate interest paid, principal paid and remaining balance, given an initial loan vs payment numbers.

Section 5: 16 numerical test cases from the calculator community as downloadable RPN programs.

The learning outcomes of Section 4 shift to understanding the TVM functions by programming same: structuring complete TVM programs and multi-solve calculations.

Problems from dm319 (Duncan's) edge case calculator tests demonstrate straightforward setup and solving and progress to teach program sequencing when multiple TVM operations are required. Problems 5/5b and 6/6b specifically teach solver bias control, showing how to obtain different valid solutions by setting initial guesses.

All programs are available in the firmware distro zip file.

- Problem 1: Monthly mortgage payment over 38 years
- Problem 1b: Reverse calculation verifying interest rate
- Problem 2: Payment differential from rate change, then PV calculation
- Problem 3: Kahan's extreme test (60×60×24×365 periods)
- Problem 4: Zero interest rate round-trip (exposes floating point precision limits)
- Problem 5: Multiple solution test (default solver finds first root at 14.44%)
- Problem 5b: Biased solver finds second root at 53.17%
- Problem 6: Multiple solution test (default finds -36.89%)
- Problem 6b: Biased solver finds second root at 58.46%
- Problem 7: Near-zero FV with fractional payments
- Problem 8: Small interest rate precision test
- Problem 9: Period calculation in END mode
- Problem 10: Period calculation in BEGIN mode
- Problem 11: Sequential FV then I/a calculation
- Problem 12: Sequential I/a then FV calculation
- Problem 12a: Sequential I/a then FV calculation

Program 'Pall' executes the complete test suite with timing, and subroutine 'TT' logs results to text files. Execution times on USB-powered C47 (DM42, DM42n) and R47 hardware range from 0.4 to 20 seconds per problem depending on the versions.

Section 1: Reference to HP TVM calculators:

Photo HP10BII+, credit

The **HP 10bII+** shown in the image on the right is referenced throughout **Section 3: Business Finance Essentials**. That tutorial was originally written with the 10bII+ as its primary example and is therefore reproduced verbatim in this application note. As a result, terminology, key names, and examples in that section follow the conventions of the 10bII+, even when equivalent operations on other calculators may be entered differently. The 10bII+ represents Hewlett-Packard's algebraic (non-RPN) financial calculator lineage. While it does not use Reverse Polish Notation, its internal time-value-of-money (TVM) model is mathematically identical to that of HP's RPN calculators, making it a useful neutral reference when explaining financial concepts independently of entry method.

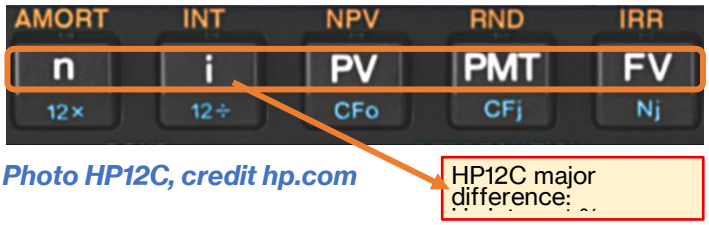


Photo HP12C, credit hp.com

The **HP-12C**, the classic workhorse of the financial industry, is similar to most modern RPN financial calculators but differs in one key respect: interest rates are entered per period rather than per year, so an annual rate must be converted explicitly at entry. While this can appear unfamiliar to users of algebraic calculators, it makes the compounding structure explicit and avoids hidden assumptions.

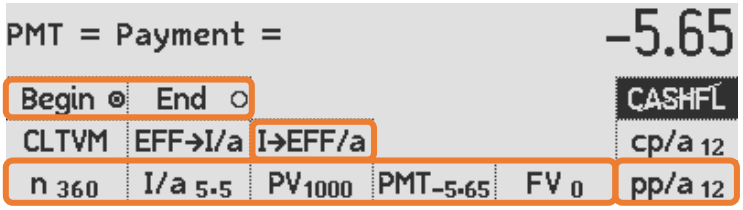
Photo HP30b, credit hp.com

The **HP 30b**, a later HP model, adds an RPN mode alongside algebraic entry, showing that differences among these calculators lie primarily in interface rather than underlying TVM mathematics. A subtle but important point is that apparent discrepancies in results almost always stem from differing conventions such as payment timing or rate entry rather than from different formulas; once these are aligned, all platforms yield the same numerical answers.

The **C47 / R47** follows the modern type, like the 30b. The functions in orange directly nod back to the HP models, where applicable:

5 Standard TVM keys:

n ⇒ The number of (payment) periods
I/a ⇒ The interest rate, I%/a (enter 10 for 10% per annum), a.k.a. discount rate or rate of return.
PV ⇒ The Present Value
PMT ⇒ The Annuity Payment
FV ⇒ The Future Value



C47 / R47 TVM menu

Special options:

EFF→I/ ⇒ Input effective rate to I%/a
I→EFF/ ⇒ Convert I%/a to effective rate /a
pp/a ⇒ Payment periods per year, generally 1 or 12.
cp/a ⇒ Compounding periods per year, generally 1 or 12.
Begin ○ End ◉ ⇒ Interest on the beginning of the period, or end of the period.

Section 2: Math Reference R47/C47 TVM:

This section gives the direct formulas used by R47/C47 in the TVM module.
The general TVM formula (3) can be used in a solver, but where possible, the analytic formulas are used. Should any formula fail due to numerical issues such as divide by zero or log of a negative number, the fallback generic equation is solved, if that might help.

1. Variable Definitions

Table of variables, as used in the TVN formulas below.

Variable	Description
PV	Present Value (principal amount today)
FV	Future Value (amount at end of term)
PMT	Payment amount per period
I%/a	Nominal annual interest rate (percentage), 5 entered for 5% per annum
NPPER	Number of payment periods (total payments)
PPER/a	Payment periods per annum (12=monthly, 4=quarterly, 1=annual)
CPER/a	Compounding periods per annum (12=monthly, 365=daily, etc.)
p	Payment timing: 0 = END mode, 1 = BEGIN mode

p is greyed out, as it is not a real accessible variable, but a R47/C47 flag, listed as ‘p’ in the formulas below.
p is controlled with the BEGIN/END buttons and the associated flag.

2. Formulas implemented in R47 and C47 (DM42n)

Nominal rate per compounding period (ic):

ic = (I%/a ÷ 100) ÷ CPER/a **(1)**

Effective interest rate per payment period (ip):

ip = (1 + ic)^(CPER/a ÷ PPER/a) − 1 **(2)**

Special case when CPER/a = PPER/a: ***ip = ic*** **(2a)**

3. General TVM Equation

0 = PV + (1+ip×p)×(PMT÷ip)×[1−(1+ip)^(−NPPER)] + FV×(1+ip)^(−NPPER) **(3)**

Special case (ip ≈ 0): 0 = PV + PMT×NPPER + FV **(3a)**

4. Solving for Present Value (PV)

PV = −[(1+ip×p)×(PMT÷ip)×[1−(1+ip)^(−NPPER)] + FV×(1+ip)^(−NPPER)] **(4)**

Special case (ip ≈ 0): PV = −PMT×NPPER − FV **(4a)**

5. Solving for Future Value (FV)

FV = −PV×(1+ip)^NPPER − (1+ip×p)×(PMT÷ip)×[(1+ip)^NPPER − 1] **(5)**

Special case (ip ≈ 0): FV = −PV − PMT×NPPER **(5a)**

6. Solving for Payment (PMT)

PMT = −[PV + FV×(1+ip)^(−NPPER)]×ip ÷ [(1+ip×p)×[1−(1+ip)^(−NPPER)]] **(6)**

Special case (ip ≈ 0): PMT = −(PV + FV) ÷ NPPER **(6a)**

7. Solving for Number of Periods (NPPER)

Case 1: No payments (PMT = 0)

NPPER = ln(−FV ÷ PV) ÷ ln(1 + ip) **(7)**

Case 2: With payments (PMT ≠ 0, ip ≠ 0)

NPPER = ln(A ÷ B) ÷ ln(1 + ip) **(8)**

A = −FV×ip + PMT×(1+ip×p) **(8a)**

B = PV×ip + PMT×(1+ip×p) **(8b)**

Case 3: Zero interest (ip ≈ 0)

NPPER = −(PV + FV) ÷ PMT **(9)**

8. Solving for Interest Rate (I%/a)

Requires iterative solution - no closed form exists: solved via the R47/C47 real solver in TVM mode

I%/a = 100×CPER/a×[(1+ip)^(PPER/a÷CPER/a) − 1] **(10)**

Where ip has to be solved for in Equation (3).

9. Amortization Schedule Calculations

Balance at start of period k (before payment):

Special case k = 0: BAL(0) = PV **(11)**

BAL(k) = BAL(k−1) × (1+ip) − PMT for k = 1, 2, ..., NPPER **(12)**

Interest paid in period k:

INT(k) = BAL(k−1) × ip **(13)**

Special case (ip ≈ 0): INT(k) = 0 **(13a)**

Principal paid in period k:

$$PRN(k) = PMT - INT(k) = PMT - BAL(k-1) \times ip \quad (14)$$

$$\text{Special case } (ip \approx 0):\ PRN(k) = PMT \quad (14a)$$

Balance after payment in period k:

$$BAL(k) = BAL(k-1) - PRN(k) = BAL(k-1) \times (1+ip) - PMT \quad (15)$$

10. Cumulative AMORT Values (periods p1 through p2)

Sum of interest paid:

$$\Sigma INT(p1 \rightarrow p2) = \Sigma(k=p1 \text{ to } p2)\ [BAL(k-1) \times ip] \quad (16)$$

Sum of principal paid:

$$\Sigma PRN(p1 \rightarrow p2) = \Sigma(k=p1 \text{ to } p2)\ [PMT - BAL(k-1) \times ip] \quad (17)$$

$$= PMT \times (p2-p1+1) - \Sigma INT(p1 \rightarrow p2) \quad (17a)$$

Remaining balance after period p2:

$$BAL(p2) = PV \times (1+ip)^{p2} - (1+ip \times p) \times (PMT \div ip) \times [(1+ip)^{p2} - 1] \quad (18)$$

$$\text{Special case } (ip \approx 0):\ BAL(p2) = PV - PMT \times p2 \quad (18a)$$

Verification (easy test) (for p1=1, p2=NPPER):

$$\Sigma INT(1 \rightarrow NPPER) + \Sigma PRIN(1 \rightarrow NPPER) = PMT \times NPPER \quad (19)$$

$$\Sigma PRIN(1 \rightarrow NPPER) = PV - BAL(NPPER) = PV - FV \quad (20)$$

Section 3: Worked out TVM tutorial Baker, Lin & Pursley

From the great tutorial[1] by Dr. Kevin Bracker; Dr. Fang Lin; and J Pursley at Pressbooks or Wayback Machine:

All references below coloured in yellow are directly taken from linked document, under the appropriate Creative Commons license NC4.0 by in this referenced document.

Example One:

6 [n] 7 [I/a] 10 000 [PV] 0 [PMT] Here, on R47/C47 you have to specify the details of compounding. Do 1 [pp/a] and [cp/a] will be defaulted to the same. Ensure End is set: [Begin ○ End ⊙]. [FV] is -15 007.30	You are investing $10,000 today and want to know how much you will have after 6 years if you earn a 7% rate of return over the 6-year time frame. Since you are starting with $10,000, that is your present value. You have 6 years, so the number of time periods is 6. The 7% rate of return means you have a 7% interest rate. In this example we are not using an annuity, so we are going to set the Annuity Payment to zero. 6 N 7 I/Y 10,000 PV 0 PMT FV -15,007.30 HP 10bII+

Example Two:

5 [n] 9 [I/a] 0 [PMT] 6 000 [FV] Here, on R47/C47 you have to specify the details of compounding. Do 1 [pp/a] and [cp/a] will be defaulted to the same. Ensure End is set: [Begin ○ End ⊙]. [PV] is -3 899.59	You are going to receive $6000 in 5 years. Assuming a 9% discount rate, what is this worth to you today? 5 N 9 I/Y 0 PMT 6000 FV PV -3899.59 HP 10bII+

Note: light grey represents settings the same as the previous problem, i.e. no need to re-enter those from the one problem to the next. Do be careful when doing the problems out of sequence.

[1] Business Finance Essentials, *TVM 5-Key Approach Guided Tutorial with HP10BII+*, Dr. Kevin Bracker; Dr. Fang Lin; and J Pursley

Example Three:

25 [n]
6 [I/a]
400 000 [PMT]
0 [FV]
Here, on R47/C47 you have to specify the details of compounding. Do 1 [pp/a] and [cp/a] will be defaulted to the same. Ensure End is set: [Begin ○ End ◉]

[PV] is -5 113 342.46

Assume you have just won a $10 Million Lottery Jackpot. However, instead of paying you the $10 Million up front, you have the choice of receiving $5 Million today or $400,000 per year at the end of each year for the next 25 years. Assuming a 6% discount rate, which would you prefer? In order to answer this, you need to find the PV of the $400,000 per year for 25 years. This is done as follows:

25 N
6 I/Y
400,000 PMT
0 FV
PV 5,113,342.46 (Note that we dropped the negative sign)

Our understanding is that the lure is 0.4M x 25 = 10M, vs. the single payment of half of that, but right now.

HP 10bII+

To get back to the original question, receiving the $10M in total over 25 years is clearly better than receiving $5M now!

Note: light grey represents settings the same as the previous problem, i.e. no need to re-enter those.

Example Three A:

All the same as Ex 3, set to BEGIN:

[Begin ◉ End ○]

[PV] is -5 420 143.01

To be more realistic, as the first 400k will be paid immediately, set to Begin, and see the 'deal' being worth even more making it the right decision to take the deal.

PV 5,420,143.01 (Note that we dropped the negative sign)

HP 10bII+

Example Three B & C: calculating using future end values

R47/C47 Solution B:

25 [n]
6 [I/a]
5 000 000 [PV]
0 [PMT]
1 [pp/a] [Begin]
[FV] is -21 459 353.60

Although the original solution THREE A (in yellow above) is the most elegant, a more "hand-waiving" intuitive way (as opposed to the quickest way) to illustrate the compounding comparison better, is equally easily done on the R47/C47:

We simply compare the tangible future values of both deals instead of the 'financial' way of evaluating the current worth of a future investment: Clearly the same answer, that $23.3M is more convincing than $21.5M:

R47/C47 Solution C:

25 [n]
6 [I/a]
0 [PV]
400 000 [PMT]
1 [pp/a] [Begin]
[FV] is -23 262 553.09

B: We first check what $5M would do for us in 25 years:

25 N
6 I/Y
5 000 000 PV
0 PMT
FV $21,459,353.60 (dropped the negative sign)

C: Then what the "deal" would do for us in the same 25 years:

25 N
6 I/Y
0 PV
400 000 PMT
FV $23,262,553.09 (dropped the negative sign)

HP 10bII+

Example Four:

20 [n] 10 [I/a] 0 [PV] 1 000 000 [FV] Here, on R47/C47 you have to specify the details of compounding. Do 1 [pp/a] and [cp/a] will be defaulted to the same. Ensure End is set: [Begin ○ End ◉]. [PMT] is -17 459.62	You want to become a millionaire and plan to do so through a savings/investment plan. Assuming you want to reach your goal in 20 years and anticipate earning a 10% rate of return, how much must you save at the end of each year in order to reach your goal? HP 10bII+ 20 N 10 I/Y 0 PV 1,000,000 FV PMT $17,459.62 This means you will need to save $17,459.62 per year in order to achieve your goal.

Note: light grey represents settings the same as the previous problem, i.e. no need to re-enter those.

Example Five:

R47/C47 has the setting Payment Periods Per Year, so type 26 [pp/a] and it is assumed the Compounding Periods Per Year [cp/a] is the same. 30 ENTER 26 X [n] 10 [I/a] (still the same as Ex 4) 0 [PV] (still the same as Ex 4) 1 000 000 [FV] (still the same as Ex 4) Ensure End is set: [Begin ○ End ◉] (still the same as Ex 4). [PMT] is -202.75	Now assume that you want to accumulate $1 million in 30 years, but instead of saving each year, you are going to save every two weeks (we will earn a 10% annual rate of return). There are 26 2-week periods in each year, so now you have to adjust your calculator to work with 26 periods per year. You can do this as follows: 26 SHIFT P/YR HP 10bII+ 780 N 10 I/Y 0 PV 1,000,000 FV PMT $202.75

Note: light grey represents settings the same as the previous problem, i.e. no need to re-enter those.

Example Six:

R47/C47 has the setting Payment Periods Per Year, so type 1 [pp/a] and it is assumed the Compounding Periods Per Year [cp/a] is the same. (still the same as Ex 4, 5) 35 [n] -3 000 [PMT] 0 [PV] (still the same form Ex 4, 5) 1 000 000 [FV] (still the same form Ex 4, 5) Ensure End is set: [Begin ○ End ◉] (still the same as Ex 4, 5). [I/a] is 10.89 %	Let's keep working with the goal of becoming a millionaire. However, instead of calculating how much you must save, we'll assume you can save $3000 per year and want to find the rate of return you will need to earn to reach your goal. This time we will give ourselves 35 years of saving $3000 per year. 35 N 0 PV -3000 PMT 1,000,000 FV I/Y 10.89% HP 10bII+

Note: light grey represents settings the same as the previous problem, i.e. no need to re-enter those.

Example Seven:

R47/C47 has the setting Payment Periods Per Year, so type 1 [pp/a] and it is assumed the Compounding Periods Per Year [cp/a] is the same. (still the same as Ex 4, 5, 6) 35 [n] (still the same as Ex 6) -3 000 [PMT] (still the same as Ex 6) -40 000 [PV] 1 000 000 [FV] (still the same as Ex 4, 5, 6) Ensure End is set: [Begin ○ End ◉] (still the same as Ex 4, 5, 6). [I/a] is 7.63 %	Here is one last variation on our millionaire example. This time, instead of starting with nothing, let's assume that we already have $40,000 and plan to save an additional $3000 per year over the next 35 years. Now, what rate of return must we earn in order to accumulate $1,000,000 at the end of the 35th year? 35 N -40,000 PV -3000 PMT 1,000,000 FV I/Y 7.63% HP 10bII+

Note: light grey represents settings the same as the previous problem, i.e. no need to re-enter those.

Examples Eight through Ten are "Cash Flow Problems" cannot yet be done by C47:
Example Eleven:

7.8 [I/a] to enter NOM% or Nominal interest value 4 [cp/a] for 4 compounding periods per year. [I→EFF/a] is 8.0311 % (Or press 4 [pp/a] for 4 Payment Periods Per Year, which does not matter, but [cp/a] will always follow.)	4 SHIFT P/YR (this sets your periods per year to 4 for quarterly compounding. Be careful here as this means all your time value of money calculations will use 4 periods per year until you change your P/YR again...just like if you changed the P/YR for a 5-key problem.) 7.8 SHIFT NOM% (this enters the 7.8% nominal rate) SHIFT EFF% (this solves for the effective annual rate to generate your final answer of 8.03%) HP 10bII+
7.6 [I/a] to enter NOM% or Nominal interest value 365 [cp/a] for 365 Periods Per Year. [I→EFF/a] is 7.8954%	For the 7.6% compounded daily it is: 365 SHIFT P/YR 7.6 SHIFT NOM% SHIFT EFF% This will give you your answer of 7.90% HP 10bII+

Section 4: AMORT tutorial

The HP 12C Platinum Loan Amortizations document from the hp.com web site is referred to below. All rights to HP, although the document claims no copyright and no license to use. All excerpts are verbatim, for educational use, and with corrections shown below in red.

Another tutorial may be found on the tvmcalc.com web site.

The examples below are from the mentioned HP 12C Platinum Loan Amortizations[2] and were checked using a web 12C calculator: https://stendec.io/ctb/rpn_fin.html.

The C47/R47 provides two AMORT calculation modes: standard mode uses mathematically exact formulas and HP12C mode which emulates HP12C's iteration method.

The HP12C rounding rounds Balance to 2 decimals after each period and uses a 10-digit periodic rate. Enable the HP12C mode via TVN AMORT menu for exact HP12C compatibility.

Example 1:
A loan is taken out for $ 180 000, at 7.75% p.a., over 30 years. The question is how much money is paid off after one year, and how much interest is paid.

Set up the relevant TVM problem: 360 [n] 7.75 [I/a] 180 000 [PV] 0 [FV] Here, on R47/C47 you have to specify the details of compounding. Do 12 [pp/a] and [cp/a] will be defaulted to the same. Ensure End is set: [Begin ○] [End ◉] [PMT] is -1 289.5420	[7] [•] [7] [5] [g] [12÷] [3] [0] [g] [12×] [1] [8] [0] [0] [0] [0] [PV] [0] [FV] [PMT] - 1,289.54 Figure 2 HP 12C Platinum
To get the interest paid and the principal paid after 12 months: Set the first period to the first month 1 [P1 1] and the last period to the 12th month [P2 12]. The total interest from month P1 = 1 to P2 = 12: press [ΣINT] is **-13 894.6655**. With HP12C ticked, **-13 894.66**. The amount of principal paid after the last month P2 = 12: press [ΣPRN] which is **-1 579.8390** and with HP12C ticked it is **-1579.82**.	[0] [n] [1] [2] [f] [AMORT] - 13,894.66 Figure 3 This is the total interest paid after one year. To see the part of the principal that is paid, press: [x⇄y] - 1,579.82 Figure 4 HP 12C Platinum
To get the amount of principal left to pay after 12 months: press [BAL] to get **178 420.1609**. With HP12C ticked it is **178 420.18**.	To check the loan balance after one year: [RCL] [PV] 178,420.18 Figure 5

[2] HP.COM: HP12C Platinum Loan Amortizations, or on the Wayback Machine Archive

Operation is not identical with the HP12C's "RCL PV", as the C47/R47 does not change the TVM solution to provide the AMORT values, which HP12C does, hence PV remain as set, at 180 000.	HP 12C Platinum

Example 2:

Example 2 continues from Example 1 with the same TVM solution in memory, and the question becomes what is the principal value and interest paid after the second year of the loan:

To move the P1 and P2 period numbers from 1-12,

HP12C □

P1 1	P2 12	ΣINT	ΣPRN	BAL	AMT_NXT

to 13-24, press [AMT_NX]:

HP12C □

P1 13	P2 24	ΣINT	ΣPRN	BAL	AMT_NXT

To get the amount of principal left to pay after 12 months:

The interest from month 13 to 24: press [ΣINT] to get **-13 767.7838**. With HP12C ticked it is **-13 767.79**.

The principal paid after month 24: press [ΣPRN] which is **-1 706.7207**. With HP12C ticked it is **-1 706.69**.

For the outstanding principal after 24 months, press [BAL]: **176 713.4403**. With HP12C ticked it is **176 713.49**.

[1] [2] [f] [AMORT]

Move 12 periods on, to 13-24

-13,767.79

Figure 7

This is the total interest paid after two years. To see the part of the principal that is paid, press:

[x⇄y]

During the second year

-1,706.69

Figure 8

To check the loan balance after one year, press:

[RCL] [PV]

after two years

176,713.49

Figure 9

HP 12C Platinum

Note: We believe the descriptive text in this HP worked problem is incorrect but not the operations.

Numerical errors: The HP tutorial answers are as per HP12-C rounding. The C47/R47 answers were checked with a manual spreadsheet and are confirmed accurate.

Examples 3 & 4 may be followed in the HP guide.

Section 5: Duncan's collection of TVM stress tests

SwissMicros Forum post of Duncan's (dm319's) problem set

From Duncan's (DM319) post, relating to comparative tests for TVM on multiple calculators. Do follow the linked reference for updated results on all tested calculators: Table from quoted link, as of 2024-09-30, modified typographically. Added text in brown.

#	Ref	N	I%YR	PV	PMT	FV	P/YR	Mode
1	DM	38 x 12	5.25%	270'000	?	0	12	end
1b	DM	38 x 12	?	270'000	-14'584/12	0	12	end
2	SlideRule	360	15% → 12%	100'000	?-?	0	12	end

HP Museum post. Calculate PMT, given n = 360, I%YR = 15%, PV = 100000, FV = 0, then calculate PMT again with I%YR = 12%. Subtract the two results, and put that back into PMT, then change n = 36 and I%YR = 15% again, and calculate PV.

3	Kahan 1983	60x60x24x365	10%	0	-0.01	?	=N	end

HP Museum post

4	DM	480	0 → ?	100'000	?→ PMT	0	12	end

Calculate PMT first given I%YR = 0, then re-input this back into PMT and calculate I%YR. On the HP-12c this is best done by pressing x<>y twice before putting back into PMT. On R47/C47 this is not required as the last solved value remains in the I/a register.

5	Dieter	10	?	50	-30	400	1	end
6	Dieter	10	?	50	-30	80	1	end

HP Museum post. Note that both have more than one solution.

7	A Chan	10	?	-100	10	1e-10	12	end

HP Museum post

8	Miguel	32	?	-999'999	0	1e6	1	end

HP Museum post

9	DM	?	25	100000	-2083.333334	0	12	end
10	DM	?	25	100000	-2040.816327	0	12	begin
11	robve	60x24x365	1/6% → ?	0	-0.01	?→ FV	=N	end
12	robve	40	? → I%YR	900	-400	-1000 → ?	1	begin

HP Museum post. Calculate for FV first, re-input back to FV and compute I%YR. 12: Calculate for i first, re-input back into i and compute FV.

Results of Duncan's TVM puzzle set as done by R47/C47, with accuracy of 34 digits

Results below from the 2026-02-23, R47/C47 version 00.109.03.01a0 pre-release, 46ea1e3:

34-digit Result R47/C47 (46ea1e3)	Reference answer
Dm319 Problem 1:	
P1 PMT = -1368.1483553505775721288281631986 80	-1368.1483553505775721288281631986801 47 [3]
Dm319 Problem 1b:	
P1b I/a = 4.3732183723100868353546878264977 51 %	4.3732183723100868353546878264977509 17 [4]
Dm319 Problem 2:	
P2 PV = 6803.0921619855971136255557644647 22	6803.0921619855971136255557644647221 25 [5]
Dm319 Problem 3:	
P3 FV = 331667.0066907768917803419084359626	331667.0066907768917803419084359625614 [6]
Dm319 Problem 4:	
P4 PMT = -208.3333333333333333333333333333333	-208.3333333333333333333333333333333333
P4 I/a = -7.9833679833679840077338876304590 08E-34 %	-7.9833679833679833679833679833679837 92E-34 [7]
Dm319 Problem 5: First solution	
P5 I/a = 14.4358713280799569742047012988770 5 %	14.4358713280799569742047012988770455 1 [8]
Dm319 Problem 5: Second solution	
P5b I/a= 53.1722132683847243104602413743711 6 %	53.1722132683847243104602413743711606 4 [9]
Dm319 Problem 6: First solution	
P6 I/a = -36.8933698741777442072399140113542 6 %	-36.8933698741777442072399140113542555 4 [10]
Dm319 Problem 6: Second solution	
P6.2 I/a= 58.4619552662209870987032074464595 8 %	58.4619552662209870987032074464595838 7 [11]
Dm319 Problem 7:	
I/a = 2.1818181818158016528925658782870 02E-10 %	2.1818181818158016528925658782870022 46E-10 [12]
Dm319 Problem 8:	
I/a = 3.1250016113292160042447454442681 75E-06 %	3.1250016113292160042447454442681746 38E-06 [13]
Dm319 Problem 9:	
n = 1060.3033900005100071273285298848 91	1060.3033900005100071273285298848906 18 [14]
Dm319 Problem 10:	
n = 1076.3195443677066464351061310443 47	1076.3195443677066464351061310443474 83 [15]
Dm319 Problem 11:	
FV = 5260.3824260003264940140903578284 63	5260.3824260003264940140903578284627 63
I%/a = 1.6666666666666666666666666666667 57E-01 %	1.6666666666666666666666666666667571 66E-01 [16]
Dm319 Problem 12a:	
I%/a = 80.0000000009845091990726031767222 0 %	80.0000000009845091990726031767222039 5
FV = -999.9999999999999999999995988604358	-999.9999999999999999999995988604358347 [17]

R47/C47 results vs. correct theoretical answers.

Red digits show where the standard R47/C47 (with 51 internal decimal digits) reach the accuracy limit. Green digits correspond with the reference values. Blue digits show 3 extra digits, so rounding can be determined. Gray digits show the rounded first part result of a 2-part problem, fixed to being a 34-digit intermediate.

The R47/C47 can achieve full theoretical accuracy when custom compiled with 75 internal decimals, which increases TVM execution times with ca. 40%. Since 75 digits benefit only the pathological case such as P4, 51 digits are chosen for production software.

---

[3] WolframAlpha Web check: WA
[4] TVM check Python3
[5] WolframAlpha Web check: WA
[6] TVM check: WA Notebook: N[(1/100) * 315360000 * ((315360001 / 315360000)^31536000 - 1), 40]
[7] The precise result is 0 but cannot be achieved due to the input rounding at 34 digits, as well as internal rounding at 51 digits
[8] TVM check Python3
[9] TVM check Python3
[10] TVM check Python3
[11] TVM check Python3
[12] TVM check Python3
[13] TVM check Python3
[14] TVM check: WA Notebook: N[-Log[1 – 1E5 * 25 / (1200 * 2083333334/1E6)] / Log[1225/1200], 40]
[15] TVM check: WA Notebook: N[-Log[1 - 2500000000000 / (2040816327 * 1225)] / Log[1225/1200], 40]
[16] WolframAlpha Web check: WA
[17] TVM check Python3

C47 & R47 relative performance on battery power, of Duncan's tests above

Performance can be measured by speed and by accuracy. The relative metrics below:

TVM tested metrics				
Model	TVM functions	Version	Speed	Accuracy
R47	ALL, Formulas, 51d, N solver	3,02a	100%	100%
C47	ALL, Formulas, 51d, N solver	3,02a	100%	100%
C47 p2	ALL, Formulas, 39d, N solver	3,02a	53%	95%
C47 p1, 3, 4	39d solver	3,02a	24%	95%

Currently available binaries with TVM performance

For good reference, the recent history of R47/C47 performance metrics are listed below with version numbers:

TVM tested metrics				
Model	TVM functions	Version	Speed	Accuracy
R47	ALL, 51d	3,01	100%	100%
C47	ALL, 51d	3,01	100%	100%
R47	Formulas	3,00	68%	88%
R47	34d solver	2,07	67%	88%
C47n	34d solver	2,07	65%	82%
C47 p4	Formulas, 39d solver	3,01	59%	95%
C47p4 *	Formulas, 39d solver	3,01	59%	99%
C47 p2	Formulas, 39d N solver	3,01	53%	95%
C47p2 *	Formulas, 51d N solver	3,01	42%	100%
C47 p3	39d N solver	3,01	29%	95%
C47	34d solver	2,07	26%	82%
C47p3 *	51d N solver	3,01	25%	100%
C47 p1	39d solver	3,01	24%	95%
C47p1 *	51d solver	3,01	19%	99%
C47	34d solver	2,00	18%	77%

C47	Older version, all function	34 digits
C47p1-4	Packaged functions	39 digits
C47p1-4 *	Special high precision test compiles	51 digits
C47 / R47	New DM42n hardware / R47 hardware	51 digits
C47 / R47	New DM42n hardware / R47 hardware	51 digits
Formulas	Explicit analytical TVM solving formulas where possible	
34d solver	Original WP43 solver, register based, 34-digit	
39d solver	39-digit Brent solver	
51d/39d N solver	Upgraded 39 / 51 digit solver using Newton-Raphson & Brent	

TVM accuracy and speed of the R47/C47 calculators

Table: Tested TVM accuracy in digits:

		C47	C47	C47n	R47	R47	C47 p1	C47 p2	C47 p3	C47 p4	C47p1 51(	C47p2 51(	C47p3 51(	C47p4 51(	C47	R47
Max		2.00b	2.07b5	2.07b5	2.07b11	3.00b0	3.01b0	3.01b0	3.01b0	3.01b0	3.01b0	3.01b0	3.01b0	3.01b0	3.01b0	3.01b0
P1	34	34	34	34	34	34	34	34	34	34	34	34	34	34	34	34
P1b	34	34	34	34	34	34	34	34	34	34	34	34	34	34	34	34
P2	34	33	33	33	33	33	34	34	34	34	34	34	34	34	34	34
P3	34	29	29	29	29	29	34	34	34	34	34	34	34	34	34	34
P4	68	19	19	19	19	19	38	39	39	38	50	49	49	50	49	49
P5	34	34	34	34	34	34	34	34	34	34	34	34	34	34	34	34
P5B	34	34	34	34	34	34	34	34	34	34	34	34	34	34	34	34
P6	34	34	34	34	34	34	34	34	34	34	34	34	34	34	34	34
P6B	34	34	34	34	34	34	34	34	34	34	34	34	34	34	34	34
P7	44	25	25	25	25	25	37	37	37	37	40	44	44	40	44	44
P8	40	37	37	37	37	37	40	40	40	40	40	40	40	40	40	40
P9	34	32	32	32	32	32	31	31	31	31	34	34	34	34	34	34
P10	34	31	31	31	31	31	31	31	31	31	34	34	34	34	34	34
P11	35		28	28	28	28	35	35	35	35	35	35	35	35	35	35
P12a	34				28	28	29	28	29	28	34	34	34	34	34	34

Number of accurate digits, example: 123000.0 has 34 digits, and 0.00123 has 34+3 digits.

Non-standard digit counting is used for comparisons:

- For n ≥ 1, counting begins at the first non-zero digit and totals 34 digits.
- For n < 1, counting begins at “0.”, and the total digit count is defined as:
  - – exponent + 34, with ‘exponent’ from the normalized scientific notation.
- Thus, 1.23E−2 = 0.0123 counts 2 + 34 digits and 1.23E−40 counts 40 + 34 digits.
- This enables meaningful comparison of small floating-point values, such as P4, whose 34 significant digits begin only at the 35th digit after the radix.

TVM operating times in seconds, on battery, for good measurement (USB power is just too fast to record). Time measurements were done internally using the LASTT command as timer:

Table: TVM operating times in seconds:

	C47 p1	C47 p2	C47 p3	C47 p4	C47p1 51d	C47p2 51d	C47p3 51d	C47p4 51d	C47	R47
Max	3.01b0	3.01b0	3.01b0	3.01b0	3.01b0	3.01b0	3.01b0	3.01b0	3.01b0	3.01b0
P1	7,9	0,1	1,0	0,1	10,5	0,1	1,5	0,1	0,0	0,0
P1b	1,4	1,6	1,8	1,2	1,6	2,0	1,8	1,6	0,8	0,7
P2	10,0	0,2	2,7	0,2	2,7	0,2	3,6	0,3	0,1	0,1
P3	2,2	0,1	2,7	0,1	18,4	0,2	3,4	0,1	0,1	0,0
P4	5,9	1,8	2,0	5,6	2,3	2,5	2,7	1,8	1,1	0,9
P5	1,2	1,6	1,7	1,0	1,2	2,1	2,3	1,5	0,9	1,0
P5B	1,3	2,3	2,3	1,5	1,8	2,8	2,8	1,7	1,2	1,2
P6	2,1	3,3	3,5	2,1	2,6	4,7	4,7	2,6	2,1	1,9
P6B	1,5	2,4	2,1	1,6	2,1	2,5	3,0	2,1	1,2	1,3
P7	1,2	0,9	0,9	1,1	1,4	1,1	1,1	1,4	0,6	0,6
P8	1,3	1,6	1,6	1,4	1,5	2,0	1,4	1,5	0,7	0,9
P9	2,5	0,1	3,0	0,1	2,9	0,1	4,1	0,2	0,1	0,1
P10	3,8	0,1	5,6	0,1	3,3	0,2	4,6	0,1	0,0	0,1
P11	1,8	2,0	3,2	0,9	2,4	1,7	2,9	1,3	0,6	0,6
P12a	2,2	2,7	3,4	1,7	2,8	3,7	4,5	2,4	1,5	1,6

Duncan's TVM problems, programmed on C47 / R47

These problems were solved using RPN programs, illustrating the use thereof on R47/C47: All programs are included in the distro as loadable programs with a state file containing all programs needed to run the tests. Run the test wit 'PAll':

Dm319 Problem 1:

N=456, i%/a=5.25, PV=270000, PMT=??, FV=0, pp/a=12, cp/a=12, END.

```
0001:  LBL 'P1'
0002:    38
0003:    12
0004:    *
0005:    STO 'NPPER'
0006:    5.25
0007:    STO 'I%/a'
0008:    270 000
0009:    STO 'PV'
0010:    0
0011:    STO 'PMT'
0012:    0
0013:    STO 'FV'
0014:    12
0015:    STO 'PPER/a'
0016:    STO 'CPER/a'
0017:    ENDP
0018:    PMT
0019:  END
```

Dm319 Problem 1b:

N=456, i%/a=??, PV=270000, PMT=-14584/12, FV=0, pp/a=12, cp/a=12, END.

```
0001:  LBL 'P1b'
0002:    38
0003:    12
0004:    *
0005:    STO 'NPPER'
0006:    0
0007:    STO 'I%/a'
0008:    270 000
0009:    STO 'PV'
0010:    -14 584
0011:    12
0012:    /
0013:    STO 'PMT'
0014:    0
0015:    STO 'FV'
0016:    12
0017:    STO 'PPER/a'
0018:    STO 'CPER/a'
0019:    ENDP
0020:    I/a
0021:  END
```

Dm319 Problem 2:

1st solve: N=360, i%/a=15, PV=100000, PMT=??, FV=0, pp/a=12, cp/a=12, END.

2nd solve: PMT solved, i%/a = 12, PMT=??

3nd solve: PMT is calculated, i%/a = 15, PV=??

```
0001:  LBL 'P2'
0002:    360
0003:    STO 'NPPER'
0004:    15
0005:    STO 'I%/a'
0006:    100 000
0007:    STO 'PV'
0008:    0
0009:    STO 'FV'
0010:    12
0011:    STO 'PPER/a'
0012:    STO 'CPER/a'
0013:    ENDP
0014:    0
0015:    STO 'PMT'     ; Bias the value needed
0016:    PMT
0017:    12
0018:    STO 'I%/a'
0019:    DROPX
0020:    PMT
0021:    -
0022:    STO 'PMT'
0023:    36
0024:    STO 'NPPER'
0025:    15
0026:    STO 'I%/a'
0027:    PV
0028:  END
```

Dm319 Problem 3:

N=31536000, i%/a=10, PV=0, PMT=-0.01, FV=??, pp/a=31536000, cp/a=31536000

```
0001:  LBL 'P3'
0002:    60
0003:    X^2
0004:    24
0005:    *
0006:    365
0007:    *
0008:    STO 'NPPER'
0009:    STO 'PPER/a'
0010:    STO 'CPER/a'
0011:    10
0012:    STO 'I%/a'
0013:    0
0014:    STO 'PV'
0015:    -0.01
0016:    STO 'PMT'
0017:    ENDP
0018:    FV
0019:  END
```

Dm319 Problem 4:

Solve 1: N=480, i%/a=0, PV=100000, PMT=??, FV=0, pp/a=12, cp/a=12, END.

Solve 2: All remains the same, I/a=?

```
0001:  LBL 'P4'
0002:    480
0003:    STO 'NPPER'
0004:    0
0005:    STO 'I%/a'
0006:    100 000
0007:    STO 'PV'
0008:    0
0009:    STO 'PMT'
0010:    STO 'FV'
0011:    12
0012:    STO 'PPER/a'
0013:    STO 'CPER/a'
0014:    ENDP
0015:    PMT
0016:    I/a
0017:  END
```

Dm319 Problem 5: First solution

Default result, without biasing the solver.

```
0001:  LBL 'P5'
0002:    10
0003:    STO 'NPPER'
0004:    0
0005:    STO 'I%/a'
0006:    50
0007:    STO 'PV'
0008:    -30
0009:    STO 'PMT'
0010:    400
0011:    STO 'FV'
0012:    1
0013:    STO 'PPER/a'
0014:    STO 'CPER/a'
0015:    ENDP
0016:    I/a
0017:  END
```

Dm319 Problem 5: Second solution

The key to the second solution is to point the solver to closer to the expected second result, i.e. store 50 in I%/a to make it search starting at 50. See brown:

```
0001:  LBL 'P5b'
0002:    10
0003:    STO 'NPPER'
0004:    0
0005:    STO 'I%/a'
0006:    50
0007:    STO 'PV'
0008:    -30
0009:    STO 'PMT'
0010:    400
0011:    STO 'FV'
0012:    1
0013:    STO 'PPER/a'
0014:    STO 'CPER/a'
0015:    ENDP
0016:    50
0017:    STO 'I%/a'
0018:    DROPX
0019:    I/a
0020:  END
```

Dm319 Problem 6: First solution

Biasing the solver to -36, see brown:

```
0001:  LBL 'P6'
0002:    10
0003:    STO 'NPPER'
0004:    0
0005:    STO 'I%/a'
0006:    50
0007:    STO 'PV'
0008:    -30
0009:    STO 'PMT'
0010:    80
0011:    STO 'FV'
0012:    1
0013:    STO 'PPER/a'
0014:    STO 'CPER/a'
0015:    ENDP
0016:    -36
0017:    STO 'I%/a'
0018:    I/a
0019:  END
```

Dm319 Problem 6: Second solution

The key to the second solution is to bias the solver to closer to the expected second result, i.e. store 50 in I%/a to make it search starting at 50. See brown:

```
0001:  LBL 'P6b'
0002:    10
0003:    STO 'NPPER'
0004:    0
0005:    STO 'I%/a'
0006:    50
0007:    STO 'PV'
0008:    -30
0009:    STO 'PMT'
0010:    80
0011:    STO 'FV'
0012:    1
0013:    STO 'PPER/a'
0014:    STO 'CPER/a'
0015:    ENDP
0016:    50
0017:    STO 'I%/a'
0018:    DROPX
0019:    I/a
0020:  END
```

Dm319 Problem 7:

```
0001:  LBL 'P7'
0002:    10
0003:    STO 'NPPER'
0004:    0
0005:    STO 'I%/a'
0006:    -100
0007:    STO 'PV'
0008:    10
0009:    STO 'PMT'
0010:    1.*10^-10
0011:    STO 'FV'
0012:    12
0013:    STO 'PPER/a'
0014:    STO 'CPER/a'
0015:    ENDP
0016:    I/a
0017:  END
```

Dm319 Problem 8:

```
0001:  LBL 'P8'
0002:    32
0003:    STO 'NPPER'
0004:    0
0005:    STO 'I%/a'
0006:    -999 999
0007:    STO 'PV'
0008:    0
0009:    STO 'PMT'
0010:    1.*10^6
0011:    STO 'FV'
0012:    1
0013:    STO 'PPER/a'
0014:    STO 'CPER/a'
0015:    ENDP
0016:    I/a
0017:  END
```

Dm319 Problem 9:

```
0001:  LBL 'P9'
0002:    0
0003:    STO 'NPPER'
0004:    25
0005:    STO 'I%/a'
0006:    100 000
0007:    STO 'PV'
0008:    -2 083.333 334
0009:    STO 'PMT'
0010:    0
0011:    STO 'FV'
0012:    12
0013:    STO 'PPER/a'
0014:    STO 'CPER/a'
0015:    ENDP
0016:    n
0017:  END
```

Dm319 Problem 10:

```
0001:  LBL 'P10'
0002:    0
0003:    STO 'NPPER'
0004:    25
0005:    STO 'I%/a'
0006:    100 000
0007:    STO 'PV'
0008:    -2 040.816 327
0009:    STO 'PMT'
0010:    0
0011:    STO 'FV'
0012:    12
0013:    STO 'PPER/a'
0014:    STO 'CPER/a'
0015:    BeginP
0016:    n
0017:  END
```

Dm319 Problem 11:

```
0001:  LBL 'P11'
0002:    REM 'Problem 11, first calculate FV then I/a'
0003:    60
0004:    24
0005:    365
0006:    *
0007:    *
0008:    STO 'NPPER'
0009:    STO 'PPER/a'
0010:    STO 'CPER/a'
0011:    1
0012:    6
0013:    /
0014:    STO 'I%/a'
0015:    0
0016:    STO 'PV'
0017:    -0.01
0018:    STO 'PMT'
0019:    ENDP
0020:    FV
0021:    I/a
0022:  END
```

Dm319 Problem 12a:

Force interest to positive root: bias 55

```
0001:  LBL 'P12a'
0002:    REM 'Problem 12 first do I/a then FV'
0003:    40
0004:    STO 'NPPER'
0005:    900
0006:    STO 'PV'
0007:    -400
0008:    STO 'PMT'
0009:    -1 000
0010:    STO 'FV'
0011:    1
0012:    STO 'PPER/a'
0013:    STO 'CPER/a'
0014:    BeginP
0015:    55
0016:    STO ‘I%/a’
0017:    I/a
0018:    FV
0019:  END
```

Sequencing program 'Pall':

Run all problem files and reporting results and time durations of processing:

```
0001:  LBL 'Pall'
0002:    0
0003:    STO 01
0004:  XEQ 'P1'
0005:    'TVM P1:'
0006:  XEQ 'TT'
0007:  XEQ 'P1b'
0008:    'TVM P1b:'
0009:  XEQ 'TT'
0010:  XEQ 'P2'
0011:    'TVM P2:'
0012:  XEQ 'TT'
0013:  XEQ 'P3'
0014:    'TVM P3:'
0015:  XEQ 'TT'
0016:  XEQ 'P4'
0017:    'TVM P4:'
0018:  XEQ 'TT'
0019:  XEQ 'P5'
0020:    'TVM P5:'
0021:  XEQ 'TT'
0022:  XEQ 'P5b'
0023:    'TVM P5B(50):'
0024:  XEQ 'TT'
0025:  XEQ 'P6'
0026:    'TVM P6:'
0027:  XEQ 'TT'
0028:  XEQ 'P6b'
0029:    'TVM P6B(50):'
0030:  XEQ 'TT'
0031:  XEQ 'P7'
0032:    'TVM P7:'
0033:  XEQ 'TT'
0034:  XEQ 'P8'
0035:    'TVM P8:'
0036:  XEQ 'TT'
0037:  XEQ 'P9'
0038:    'TVM P9:'
0039:  XEQ 'TT'
0040:  XEQ 'P10'
0041:    'TVM P10:'
0042:  XEQ 'TT'
0043:  XEQ 'P11'
0044:    'TVM P11:'
0045:  XEQ 'TT'
0046:  XEQ 'P12a'
0047:    'TVM P12a:'
0048:  XEQ 'TT'
0049:  END
```

Subroutine 'TT':

Print results to text files:

```
0001:  LBL 'TT'
0002:    LASTT
0003:    STO 00
0004:    DROPX
0005:    Printxy
0006:    RCL 00
0007:    RCL 01
0008:    -
0009:    10.0
0010:    /
0011:    ' Sec'
0012:    SIG 02
0013:    +
0014:    Printx
0015:    RCL 00
0016:    STO 01
0017:  END
```

Original	012:	2024-06-01, on C47 version 00.109.02.00
Updated	012a:	2025-04-22, on C47 version 00.109.00.07b5
Updated	012b:	2026-02-05, on R47/C47 version 00.109.03.00
Updated	012c:	2026-02-22, on R47/C47 version 00.109.03.01
Updated	012d:	2026-02-25, on R47/C47 version 00.109.03.01

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